

Group Presentation: Numerical Methods for Population Growth Analysis

CASE STUDY 1: Population Growth Analysis Using Numerical Methods

Scenario

A local government wants to analyze how population changes over time. However, they only have yearly population data, not a continuous function.

Students must:

- Estimate population growth rate (derivative)
- Estimate total population change (integration)
- Analyze trends

Given Data:

Year	Population
2020	10,000
2021	10,800
2022	11,900
2023	13,200
2024	14,800

1. Numerical Differentiation (Growth Rate)

Formula Used: Central Difference

$$P'(t) = \frac{P(t+1) - P(t-1)}{2}$$

Compute growth rate at:

t = 1, 2, 3

- **t = 1(2021)**

$$P'(1) = \frac{11900 - 10000}{2} = 950$$

- **t = 2(2022)**

$$P'(2) = \frac{13200 - 10800}{2} = 1200$$

- $t = 3(2023)$

$$P'(3) = \frac{14800 - 11900}{2} = 1450$$

Growth Rate Table:

Year	Population	Growth Rate (people/year)
2020	10,000	-
2021	10,800	950
2022	11,900	1200
2023	13,200	1450
2024	14,800	-

2. Numerical Integration (Total Change)

Use Trapezoidal Rule to estimate:

$$\int_{2020}^{2024} P(t) dt$$

$$\int_{2020}^{2024} P(t) dt \approx \sum_i \frac{P(t_i) + P(t_{i+1})}{2} \cdot \Delta t$$

Interval	$P(t_i)$	$P(t_{i+1})$	Trapezoid Area
2020 - 2021	10,000	10,800	10,400
2021 - 2022	10,800	11,900	11,350
2022 - 2023	11,900	13,200	12,550
2023 - 2024	13,200	14,800	14,000

$$2020 - 2021: \frac{10,000 + 10,800}{2} = 10,400$$

$$2021 - 2022: \frac{10,800 + 11,900}{2} = 11,350$$

$$2022 - 2023: \frac{11,900 + 13,200}{2} = 12,550$$

$$2023 - 2024: \frac{13,200 + 14,800}{2} = 14,000$$

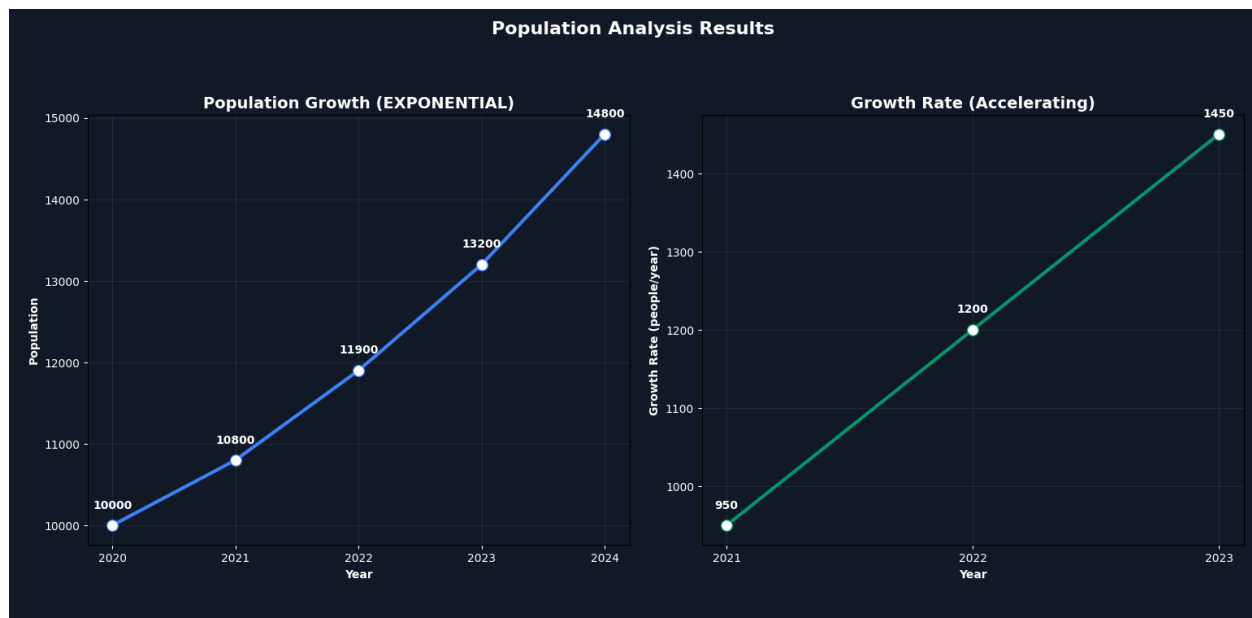
Total = 10,400 + 11,350 + 12,550 + 14,000

Total = 48,300 person-years

3. Visualization

- Plot:

- Population vs time
- Growth rate vs time



4. Analysis Questions

- When did growth accelerate most?

- The entire duration (the growth rate increases steadily by 250 people/year annually).

- Is the growth linear or exponential?

- Exponential (the growth rate is increasing over time, not constant).

- Predict population in 2025

- Approximately 16,500 (based on adding the current growth rate of 1,450 plus the consistent acceleration of 250).

Key Findings

1. Growth Rate Trends

Numerical differentiation using the Central Difference formula revealed that the population is not growing at a constant rate, but rather at an accelerating pace.

- **Calculated Rates:** The growth rate increased from 950 people/year in 2021 to 1,450 people/year in 2023.
- **Constant Acceleration:** There is a consistent annual acceleration of 250 people/year in the growth rate.

2. Cumulative Population Impact

Numerical integration via the Trapezoidal Rule was utilized to calculate the total population change over the study period.

- **Total Estimate:** The integration resulted in a value of **48,300 person-years**. This metric represents the cumulative population exposure over the analyzed interval.

3. Trend Classification and Future Outlook

The data indicates that the population growth is **exponential** rather than linear, as evidenced by the increasing growth rate over time.

- **Prediction:** Based on the identified pattern of consistent acceleration, the estimated population for the year 2025 is approximately **16,500**.